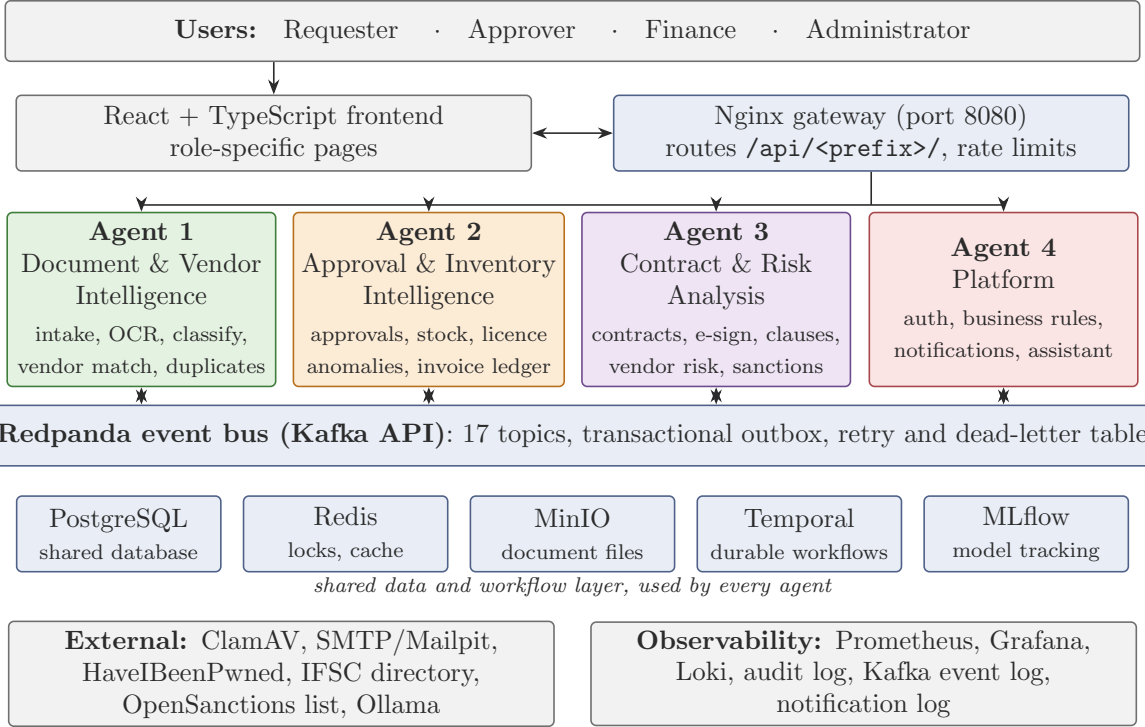

Users: Requester · Approver · Finance · Administrator
React + TypeScript frontend
role-specific pages
Nginx gateway (port 8080)
routes /api/<prefix>/, rate limits
Agent 1
Document & Vendor Intelligence
intake, OCR, classify, vendor match, duplicates
Agent 2
Approval & Inventory Intelligence
approvals, stock, licence anomalies, invoice ledger
Agent 3
Contract & Risk Analysis
contracts, e-sign, clauses, vendor risk, sanctions
Agent 4
Platform
auth, business rules, notifications, assistant
Redpanda event bus (Kafka API): 17 topics, transactional outbox, retry and dead-letter table
PostgreSQL
shared database
Redis
locks, cache
MinIO
document files
Temporal
durable workflows
MLflow
model tracking
shared data and workflow layer, used by every agent
External: ClamAV, SMTP/Mailpit, HaveIBeenPwned, IFSC directory, OpenSanctions list, Ollama
Observability: Prometheus, Grafana, Loki, audit log, Kafka event log, notification log